The Central Limit Theorem

STAT 200 – Introductory Statistics

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Today's Objective

After today's lecture, you should:

- State the Central Limit Theorem.
- State the requirements for applying it.
- State its consequences:
  - with respect to the distribution of the sample mean.
  - with respect to the distribution of the sample proportion.

The Central Limit Theorem

The Central Limit Theorem

Let X be a random variable with mean μ and finite variance σ^2. Let us draw a random sample of size n from this distribution.

Then, the distribution of the sample sums converges to a Normal distribution as n gets larger. Specifically,

$$\sum_{i=1}^{n} X_i \xrightarrow{d} \textsc{Normal}(n\mu,\, n\sigma^2).$$

The proof of this theorem is beyond the scope of this course. It is first proven in MATH 321: Mathematical Statistics I.

Consequences of the Central Limit Theorem

The Central Limit Theorem (CLT) tells us the following:

- The *sum* of independent random variables is more NORMAL than the distribution of the variable itself, unless the variable is NORMAL distributed or if it has an infinite variance.
    - The BINOMIAL is a sum of independent BERNOULLI random variables.
    - The POISSON is a sum of independent POISSON random variables.
- Because the sample mean is just the sample sum, divided by a constant (n), the Central Limit Theorem tells us that *the distribution of sample means* will tend towards the NORMAL distribution.
- The speed of convergence depends on how closely the data distribution is to a NORMAL distribution. The closer, the faster.

Example #1: The UNIFORM Distribution

Example

Let $X \sim \textsc{Uniform}(a, b)$. Use the Central Limit Theorem to estimate the distribution of the sum of a sample of size n.

By the CLT,

$$T \sim \textsc{Normal}(n\mu, n\sigma^2).$$

From our knowledge of the UNIFORM distribution, this means

$$T \sim \textsc{Normal}\left(n\frac{a+b}{2}, n\frac{(b-a)^2}{12}\right)$$

Example #2: The Exponential Distribution

Example

Let $X \sim \text{Exponential}(\lambda)$. Use the Central Limit Theorem to estimate the distribution of the sum of a sample of size n.

By the CLT,

$$T \sim \text{Normal}(n\mu, n\sigma^2).$$

From our knowledge of the Exponential distribution, this means

$$T \sim \text{Normal}\left(n\frac{1}{\lambda}, n\frac{1}{\lambda^2}\right)$$

Example #3: The BINOMIAL Distribution

Example

Let $X \sim \text{BINOMIAL}(n, p)$. Use the Central Limit Theorem to approximate the distribution of X.

Note that the BINOMIAL is just the sum of n independent BERNOULLI distributions. That is, if $Y_i \sim \text{BERNOULLI}(p)$m then $Y_i \sim \text{BINOMIAL}(1, p)$ and,

$$X = \sum_{i=1}^{n} Y_i \sim \text{BINOMIAL}(n, p).$$

Thus, by the Central Limit Theorem, we know

$$X \sim \text{NORMAL}(np, np(1-p)).$$

The Distribution of the Sample Mean

Let X be a random variable with mean μ and finite variance σ^2. Let us draw a random sample of size n from this distribution.

Then, the distribution of the sample mean is

$$\overline{X}_n \xrightarrow{d} \textsc{Normal}\left(\mu,\, \frac{1}{n}\sigma^2\right)$$

Proof of the Distribution of the Sample Mean

Proof:

From the Central Limit Theorem, we know

$$\sum_{i=1}^{n} X_i \sim \textsc{Normal}(n\mu, n\sigma^2).$$

Thus,

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \sim \textsc{Normal}\left(\mu, \frac{1}{n}\sigma^2\right)$$

Proof of the Distribution of the Sample Mean

Proof of Expectation:

$$\begin{aligned} \mathbb{E}\left[\,\overline{X}_n\,\right] &= \mathbb{E}\left[\frac{1}{n}T\right] \\ &= \frac{1}{n}\mathbb{E}\left[\,T\,\right] \\ &= \frac{1}{n}\left(n\mu\right) \\ &= \mu \end{aligned}$$

Proof of the Distribution of the Sample Mean

Proof of Variance:

$$\begin{aligned}
\mathbb{V}\left[\,\overline{X}_n\,\right] &= \mathbb{V}\left[\frac{1}{n}T\right] \\
&= \frac{1}{n^2}\mathbb{V}\left[\,T\,\right] \\
&= \frac{1}{n^2}\left(n\sigma^2\right) \\
&= \frac{1}{n}\sigma^2
\end{aligned}$$

NOTE: Sample Means vs. Sample Values

Before we move onto examples, I want to make an important distinction. The Central Limit Theorem describes the behavior of the ***sample mean***, not the ***sample values***!

The sample values will stay in whatever distribution the population takes. The sample means will approach Normality.

It is a subtle difference, but a very important one.

`https://www.statcrunch.com/applets/type3&samplingdist`

Example #1: A First Look

Example

I draw a sample of size $n = 14$ from a population with mean $\mathbb{E}[X] = 126$ and variance $\mathbb{V}[X] = 42$. What is the approximate distribution of the sample means?

Example #2: Heights

Example

I have been told that the average adult height for males in the United States has mean $\mu = 69$ inches and standard deviation $\sigma = 3$ inches. What is the probability of having the mean of a sample of size 2 being less than 65 inches?

Example #3: More Heights

Example

I have been told that the average adult height for males in the United States has mean $\mu = 69$ inches and standard deviation $\sigma = 3$ inches. What is the probability of having the mean of a sample of size 10 being less than 65 inches?

Example #4: Crime

Example

The 2000 violent crime rate for the 50 states (and the District of Columbia) are given in the data file `crime`. What is a 95% central confidence interval for the mean violent crime rate?

Example #4: Crime (Bootstrapping)

We could also estimate the confidence interval from the data. This process is called "bootstrapping", and here is the code:

```
mn = numeric()
for (i in 1:1e4) {
    x = sample(vcrime00, replace = TRUE)
    mn[i] = mean (x)
}
quantile(mn, c(0.025, 0.975))
```

This gives a 95% confidence interval of 380 to 510.

Question: Why is there a difference between the two confidence intervals?

The Distribution of the Sample Proportion

The Distribution of the Sample Proportion

Let $X \sim \textsc{Binomial}(n, p)$ be a random sample of size n from Bernoulli random variables.

Then, the distribution of the sample proportion is

$$P = \frac{1}{n} X \xrightarrow{d} \textsc{Normal}\left(p, \frac{p(1-p)}{n}\right)$$

Proof of the Distribution of the Sample Proportion

Proof:

From the Central Limit Theorem, we know

$$X \sim \textsc{Normal}(np, np(1-p)).$$

Thus,

$$\mathbb{E}[P] = \mathbb{E}\left[\frac{X}{n}\right] = \frac{\mathbb{E}[X]}{n} = \frac{np}{n} = p,$$

$$\mathbb{V}[P] = \mathbb{V}\left[\frac{X}{n}\right] = \frac{\mathbb{V}[X]}{n^2} = \frac{np(1-p)}{n^2} = \frac{p(1-p)}{n},$$

and

$$P \sim \textsc{Normal}\left(p, \frac{p(1-p)}{n}\right).$$

Example #1: Poverty

Example

According to the US Census, 18% of Americans are below the poverty line. If I randomly sample $n = 10$ people from the United States, what is the probability that more than 20% of them are below the poverty line?

Example #2: More Poverty

Example

According to the US Census, 18% of Americans are below the poverty line. If I randomly sample $n = 100$ people from the United States, what is the probability that more than 20% of them are below the poverty line?

Example #3: Much More Poverty

Example

According to the US Census, 18% of Americans are below the poverty line. If I randomly sample $n = 1000$ people from the United States, what is the probability that more than 20% of them are below the poverty line?

Example #4: Even More Poverty

Example

According to the US Census, 18% of Americans are below the poverty line. If I randomly sample $n = 10000$ people from the United States, what is the probability that more than 20% of them are below the poverty line?